

FinPilot - Study Guide (12-Hour Preparation)

Project: FinPilot - Financial Intelligence & Tax Management Platform

Stack: React * Django * PostgreSQL * MongoDB * Scikit-learn

Goal:

Learn the complete project in one day for project submission and viva.

Study Order

Project Overview

->

Folder Structure

->

Authentication

->

Clients

->

Companies

->

Transactions

->

Documents

->

Tax

->

Machine Learning

->

Analytics

->

Database

->

Complete Demo

->

Viva

Phase 1 - Project Overview (1 Hour)

Learn

- What problem does FinPilot solve?

- Why was it built?
- Who will use it?
- What are the main features?
- Why React?
- Why Django?
- Why MongoDB?
- Why PostgreSQL?
- Why Machine Learning?

Read

```
PROJECT_PRESENTATION.md
VIVA_PREPARATION.md
FINAL_SUBMISSION_REPORT.md
```

Understand

User

->

Frontend

->

Backend

->

Database

->

Response

->

Frontend

Phase 2 - Folder Structure (30 Minutes)

Draw this from memory.

Frontend

```
frontend/src/
```

```
components/  
context/  
features/  
routes/  
lib/
```

Features

```
auth/  
clients/  
companies/  
transactions/  
documents/  
tax/  
analytics/  
ml/  
dashboard/  
profile/  
system/  
landing/  
errors/
```

Know one sentence for each folder.

Backend

```
backend/apps/  
  
accounts/  
clients/  
companies/  
transactions/  
documents/  
tax/  
analytics/  
ml/  
dashboard/  
system/  
mongo/
```

Know one sentence for each module.

Phase 3 - Authentication (1 Hour)

Backend Files

backend/apps/accounts/views.py

backend/apps/accounts/serializers.py

backend/apps/accounts/models.py

backend/apps/accounts/permissions.py

backend/apps/accounts/urls.py

Frontend Files

frontend/src/features/auth/

AuthContext.jsx

api.js

authorization.js

pages/

LoginPage.jsx

RegisterPage.jsx

VerifyOtpPage.jsx

ForgotPasswordPage.jsx

ResetPasswordPage.jsx

Shared

frontend/src/lib/axios.js

frontend/src/components/ProtectedRoute.jsx

frontend/src/routes/router.jsx

Learn

Registration

->

OTP Verification

->

Login

->

JWT

->

Refresh Token

->

Logout

->

Forgot Password

Phase 4 - Clients (30 Minutes)

Backend

```
backend/apps/clients/
```

```
views.py
```

```
services.py
```

```
serializers.py
```

```
urls.py
```

Frontend

```
frontend/src/features/clients/
```

```
api.js
```

```
hooks.js
```

```
pages/
```

```
ClientListPage.jsx
```

```
ClientDetailPage.jsx
```

Learn

Create Client

->

Update Client

->

Delete Client

->

Map Client to Companies

Phase 5 - Companies (45 Minutes)

Backend

```
backend/apps/companies/
```

```
views.py
```

```
services.py
```

```
serializers.py
```

```
urls.py
```

Frontend

```
frontend/src/features/companies/
```

```
api.js
```

```
hooks.js
```

```
pages/
```

```
CompanyListPage.jsx
```

```
CompanyDetailPage.jsx
```

Learn

Create Company

->

Update Company

->

Delete Company

->

Select Active Company

Phase 6 - Transactions (2 Hours)

Backend

backend/apps/transactions/

views.py

services.py

serializers.py

urls.py

Frontend

frontend/src/features/transactions/

api.js

hooks.js

pages/

TransactionsPage.jsx

CategoriesPage.jsx

VendorsPage.jsx

CustomersPage.jsx

Learn

Transaction CRUD

->

Categories

->

Vendor

->

Customer

->

Budget

->

CSV Import

->

Excel Import

->

Export

->

Analytics Data

Ignore helper functions.

Phase 7 - Documents (45 Minutes)

Backend

```
backend/apps/documents/
```

```
views.py
```

```
services.py
```

```
serializers.py
```

Frontend

```
frontend/src/features/documents/
```

```
DocumentsPage.jsx
```

```
DocumentDetailPage.jsx
```

Learn

Upload

->

Store

->

Duplicate Detection

->

Archive

Phase 8 - Tax (45 Minutes)

Backend

backend/apps/tax/

views.py

services.py

gst_rules.py

tds_rules.py

Frontend

frontend/src/features/tax/

TaxCenterPage.jsx

Learn

GST

->

TDS

->

Income Tax

->

Compliance

->

Filing Readiness

Phase 9 - Machine Learning (45 Minutes)

Backend

backend/apps/ml/

train.py

predict.py

features.py

services.py

utils.py

Frontend

frontend/src/features/ml/

AIDashboardPage.jsx

Learn

Expense Categorization

->

Random Forest

Duplicate Detection

->

Isolation Forest

Compliance Risk

->

Decision Tree

Expense Forecast

->

Linear Regression

Tax Forecast

->

Linear Regression

Don't memorize algorithms.

Only know why each one is used.

Phase 10 - Analytics (45 Minutes)

Backend

```
backend/apps/analytics/
```

```
views.py
```

```
services.py
```

```
utils.py
```

Frontend

```
frontend/src/features/analytics/
```

```
AnalyticsDashboardPage.jsx
```

```
ExecutiveDashboardPage.jsx
```

```
ReportCenterPage.jsx
```

Learn

KPIs

->

Mongo Aggregation

->

Charts

->

Reports

->

Export PDF

->

Export Excel

Phase 11 - Dashboard (30 Minutes)

Backend

```
backend/apps/dashboard/
```

views.py

services.py

Frontend

DashboardPage.jsx

Learn

Dashboard

->

Statistics

->

Recent Activity

->

Quick Actions

Phase 12 - Routing (30 Minutes)

Read

frontend/src/routes/router.jsx

Understand

Landing

->

Login

->

Dashboard

->

Clients

->

Companies

->

Transactions

->

Tax

->

Analytics

->

ML

->

Settings

->

Logout

Phase 13 - API Flow (1 Hour)

Example

TransactionsPage.jsx

->

hooks.js

->

api.js

->

Axios

->

Django View

->

Service

->

MongoDB

->

Response

->

React Query

->

UI

If you understand one flow,
you understand almost the whole project.

Phase 14 - Database (45 Minutes)

PostgreSQL

Know these tables

Users

EmailVerificationOTP

PasswordResetOTP

OutstandingToken

BlacklistedToken

AuditLog

Purpose

Authentication

->

OTP

->

JWT

->

Audit

MongoDB

Know these collections

```
clients
client_company_mapping
company_members
companies
transactions
documents
transaction_categories
vendors
customers
parties
budgets
analytics
reports
analytics_report_log
ml_models
user_preferences
```

Purpose

Business Data

Phase 15 - Run Complete Demo (2 Hours)

Repeat this until comfortable.

Landing

->

Register

->

Verify OTP

->

Login

->

Create Client

->

Create Company

->

Add Vendor

->

Add Customer

->

Create Transaction

->

Import CSV

->

Upload Documents

->

Tax Center

->

Analytics

->

ML Dashboard

->

Train Models

->

Download Report

->

Profile

->

Settings

->

Logout

Phase 16 - Viva Preparation (1 Hour)

For every module answer:

1

What does it do?

2

Which files implement it?

3

How does data flow?

Example

Transactions

Purpose

Manage financial records.

Backend

views.py

services.py

Frontend

TransactionsPage.jsx

api.js

hooks.js

Database

MongoDB

transactions collection

Important Files

Frontend

routes/router.jsx

AuthContext.jsx

ProtectedRoute.jsx

DashboardLayout.jsx

ThemeContext.jsx

lib/axios.js

features/clients/api.js

features/companies/api.js

features/transactions/api.js

ClientListPage.jsx

CompanyListPage.jsx

TransactionsPage.jsx

Backend

config/settings.py

config/urls.py

accounts/views.py

clients/views.py

companies/views.py

transactions/views.py

documents/views.py

tax/views.py

analytics/views.py

ml/train.py

ml/predict.py

Don't Waste Time Learning

Skip these unless asked.

Utility files

Every helper

Every regex

Every serializer field

Tailwind implementation

Every React hook

Every import statement

Small helper functions

Understand the project.

Don't memorize code.

12-Hour Timeline

Time	Task
-----	-----
1 Hour	Project Overview
30 min	Folder Structure
1 Hour	Authentication
30 min	Clients
45 min	Companies
2 Hours	Transactions
45 min	Documents
45 min	Tax
45 min	Machine Learning
45 min	Analytics
45 min	Database
2 Hours	Demo Practice
1 Hour	Viva Practice

Final Goal

By the end of this guide, you should be able to explain:

- Overall Architecture
- Authentication Flow
- React Frontend
- Django Backend
- PostgreSQL
- MongoDB
- Machine Learning
- Tax Module
- Analytics
- Complete User Flow

without needing to memorize every line of code.